Modeling the disaster relief hub network design problem: A case of Central Vietnam

Tang-Phu-Quy Le[1], Quang-Vu Nguyen[1], and Sunarin Chanta[2*]

[1] The University of Danang, Vietnam - Korea University of Information and Communication Technology
[2] King Mongkut's University of Technology North Bangkok
quyltp.22git@vku.udn.vn, nqvu@vku.udn.vn, sunarin.c@itm.kmutnb.ac.th

Abstract. Natural disasters severely disrupt transportation infrastructure, heavily complicating search and rescue (SAR) operations. Existing humanitarian logistics models often fail to capture these dynamic disruptions and the non-linear escalation of human suffering. To address this gap, we propose a two-stage stochastic Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP). Our model simultaneously minimizes expected logistics costs – utilizing Daganzo's Continuous Approximation for scalable last-mile routing – and the expected maximum deprivation cost to enforce social equity. To solve this NP-hard problem, we develop a Priority-Based Non-dominated Sorting Genetic Algorithm II (PB-NSGA-II). Extensive computational experiments on standard benchmarks (AP, TR81) validate the algorithm's efficiency. Furthermore, a detailed case study of the flood-prone Vu Gia – Thu Bon river basin provides crucial managerial insights. The results highlight the necessity of dynamically adapting multi-modal network topologies to maintain resilience under extreme disaster scenarios.

Keywords: Humanitarian Logistics · Disaster relief · Hub location problem · Multi-objective Evolutionary Algorithm.

1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [2, 4]. As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [12, 15].

There were many great challenges in humanitarian operations during this period, one of them being the severe disruption of national logistics infrastructure. The North-South Railway – backbone of Vietnam's transport network was

* Corresponding author.

paralyzed due to track erosion and submersion, forcing the cancellation of 170 trains [12]. Simultaneously, key arteries such as the National Highway 1 and routes in the Central Highlands were severed by landslides, isolating communities for days [10]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed. The physical network becomes "incomplete", characterized by broken links and isolated nodes. Another challenge is resource scarcity resulting from poor planning and uncertain supply and demand. Disaster management involves four phases: mitigation, preparedness, response, and recovery [1]. Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing.

1.2 Related Works

The design of disaster relief networks differs fundamentally from commercial logistics. Commercial models prioritize profit maximization. Humanitarian logistics prioritizes human lives and rapid response [14]. Facility location is a critical component of disaster preparedness. A comprehensive review by Boonmee et al. [3] highlights the growing need for facility location models under severe uncertainty.

Disruption and Network Design. Disasters often cause severe infrastructure disruptions. Standard network models fail when paths become unavailable. Yahyaei and Bozorgi-Amiri [16] addressed this by proposing a robust reliable network design. They integrated shelter and supply facility locations under disruption risks. Similarly, Hasani and Mokhtari [8] proposed an integrated relief network model for Iran. They modeled dynamic uncertainty and employed multi-objective optimization. However, embedding exact routing sequences into these network designs creates highly intractable Location-Routing Problems (LRP).

Continuous Approximation. To circumvent the NP-hard complexity of exact routing, researchers utilize approximations. Daganzo [5] pioneered the Continuous Approximation (CA) method. CA estimates last-mile routing costs using spatial densities rather than explicit node-to-node sequences. This technique drastically reduces computational burden. It allows large-scale network models to remain tractable. Our model adopts this CA framework to estimate evacuation costs efficiently.

Deprivation Cost and Social Equity. Traditional objective functions usually minimize pure logistics costs or average travel times. Holguín-Veras et al. [9] argued that linear time penalties neglect social equity. They introduced the concept of deprivation cost. It applies an exponential penalty function to waiting times. This mathematically captures the non-linear escalation of human suffering during disasters. Incorporating deprivation cost forces optimization models to distribute resources equitably. It prevents the marginalization of isolated distress clusters.

Solution Approaches. Disaster relief models often involve multiple conflicting objectives. Meta-heuristics are highly preferred for these problems. NSGA-II [6] is a standard and robust evolutionary framework. It utilizes non-dominated sorting and crowding distance to find Pareto-optimal solutions. Recent studies frequently adapt NSGA-II for humanitarian operations.

Research Gap and Our Contribution. Existing literature contains a significant gap. Very few studies integrate incomplete hub networks, Daganzo's continuous approximation, and exponential deprivation costs under a two-stage stochastic framework. Solving such a combined model is computationally prohibitive for exact solvers. Our study addresses this gap. We formulate the MO-IHLNDP and propose a meta-heuristic algorithm named Priority-Based NSGA-II, which can handle large-scale instances effectively.

2 Problem Description and Mathematical Model

To design an efficient disaster relief network (DRND), specifically adjusted for flood disaster response in Central Vietnam, we framed the problem as a multi-objective incomplete hub location network design problem (MO-IHLNDP). This model is the capacitated, single-allocation variant of the HLP.

The main components of our network are origins, hubs, and demands. In our practical settings, hubs refer to the rescue stations. There are two kinds of hubs, corresponding to two kinds of strategies. The first type is established in advance of the disasters, strategically positioned, and inventoried with amenities. The second type is opened dynamically for disaster response and has high approachability to affected areas. Lastly, during the disaster response, we gradually discover the victim groups, which are referred to as **demands**. Besides the government's planned forces, some volunteers spontaneously offer to help; these are referred to as **origins**.

From a practical point of view, DRND involves multiple critical uncertainty factors. Firstly, the disruption of transportation infrastructure (e.g., damaged roads and buildings) can easily disable pre-programmed plans, urging the need for a dynamic solution [16]. Secondly, the exact locations of the demands, the total number of victims, and the availability of supply origins are highly uncertain during the rescue process [13].

To account for these uncertainties, we employ a two-stage stochastic programming approach utilizing multiple scenarios. In each scenario, a specific risk factor is assigned to each area. We also assume that search and rescue (SAR) operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy the uncertain demands under extreme scenarios.

Finally, the model places a strong emphasis on *social equity* by integrating the deprivation cost [9]. To obtain a robust approximation of the rescue time from an established hub to its assigned demand points without causing computational intractability, we utilize Daganzo's continuous approximation formula [5], a technique widely adopted in routing problems.

2.1 Sets and Indices

$\mathcal{H}, \mathcal{I}, \mathcal{J}$	Set of candidate hubs (k, h), demand nodes (i), and origins (j)
$\mathcal{S}, \mathcal{M}$	Set of disaster scenarios (s) and transportation modes (m)
$\mathcal{V}$	Set of all nodes (u, v) representing the area in question, $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$

2.2 Parameters

First-Phase Parameters (Strategic Planning)

F_k	Fixed cost to establish hub k of the first type (before the disaster)
c_k, κ_k	Unit holding cost and capacity limit for inventory at hub k
α, χ, M	Discount factor for economies of scale, max acceptable risk threshold, and Big-M constant
C_{uvm}, τ_{uvm}	Unit transportation cost and estimated travel time from node u to v via mode m
C_m, Q_m	Unit local routing cost and transportation capacity for vehicle mode m
γ	Conversion factor (average amount of relief items required per evacuated person)

Second-Phase Parameters (Disaster Response, Scenario s)

π_s	Probability of scenario s
F^a_{ks}	Fixed cost to establish hub k of the second type (reactive) in scenario s
τ_{ks}	Processing time for each rescue round trip at hub k in scenario s
a_{uvms}	Accessibility of arc (u, v) via mode m in scenario s (1 if traversable, 0 otherwise)
r_{us}, λ_{is}	Risk index of node u, and deprivation sensitivity coefficient for demand i in scenario s
D_{is}, O_{js}	Demand (number of people to evacuate) at i, and relief items provided at origin j in scenario s
Θ_{kis}	Pre-computed CA routing cost to execute the evacuation from grid i to hub k in scenario s
Φ, η, A_i	Daganzo's circuity factor, average group size per distress location, and area of grid cell i

2.3 Decision Variables

$x_k \in \{0, 1\}$	1 if hub k is established as the first type (planned), 0 otherwise
$y_{ks} \in \{0, 1\}$	1 if hub k is established as the second type (reactive) in scenario s, 0 otherwise
$z_{iks} \in \{0, 1\}$	1 if demand i is assigned to be evacuated to hub k in scenario s, 0 otherwise
$z_{jks} \in \{0, 1\}$	1 if origin j supplies hub k in scenario s, 0 otherwise
$q_k \geq 0$	Amount of relief items inventoried at hub k
$f_{khms} \geq 0$	Lateral trans-shipment flow volume from hub k to hub h via mode m in scenario s

2.4 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$\begin{aligned}
\text{Min } Z_1 = & \underbrace{\sum_{k \in \mathcal{H}} F_k x_k + \sum_{k \in \mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\
& + \sum_{s \in \mathcal{S}} \pi_s \Bigg[\underbrace{\sum_{k \in \mathcal{H}} F^a_{ks} y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j \in \mathcal{J}} \sum_{k \in \mathcal{H}} \left(\min_{m \in \mathcal{M} | a_{jkms}=1} C_{jkm} \right) O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \\
& + \underbrace{\sum_{k \in \mathcal{H}} \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k \in \mathcal{H}} \sum_{i \in \mathcal{I}} \Theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \Bigg]
\end{aligned} \tag{1}$$

Objective Z_1 minimizes the total expected logistics cost, balancing pre-disaster investments against reactive setups and transportation costs. To have an accurate estimation of last-mile evacuation cost while keeping the model simple, we adopted Daganzo's Continuous Approximation [5]. The region of interest is partitioned via spatial discretization, and the demands are aggregated on a grid basis. The routing cost Θ_{kis} is pre-computed as a parameter:

$$\Theta_{kis} = \min_{m \in \mathcal{M}} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\text{Min } Z_2 = \sum_{s \in \mathcal{S}} \pi_s \left(\max_{i \in \mathcal{I}} \left[D_{is} \cdot \left(e^{\lambda_{is} \cdot \Omega_{is}} - 1 \right) \right] \right) \tag{2}$$

The second objective ensures social equity by minimizing the expected maximum deprivation cost across all scenarios. Based on Holguín-Veras et al. [9], Z_2 adopts an exponential function to penalize long waiting times (Ω_{is}) for victims. Furthermore, it forces the model to distribute resources evenly, ensuring overall low delay time for everyone.

$$\Omega_{is} = \sum_{k \in \mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m \in \mathcal{M}} \tau_{kim} \right)$$

The sensitivity coefficient λ_{is} is proportional to the risk index r_{is} ($\lambda_{is} = \lambda_0(1 + r_{is})$), effectively prioritizing vulnerable populations. Although Z_2 introduces non-linearity, our meta-heuristic solution approach can directly evaluate it as a fitness function. For linearization into a Mixed-Integer Linear Programming (MILP) form, one can exploit the single-allocation property ($\sum_{k \in \mathcal{H}} z_{iks} = 1$).

2.5 Constraints

The complete MO-IHLNDP model is formally stated as follows:

$$\begin{aligned}
&\text{Minimize} \quad Z_1, Z_2 \\
&\text{subject to} \quad \text{Constraints } (3) - (16)
\end{aligned}$$

$$x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{3}$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \tag{4}$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{5}$$

$$\sum_{k \in \mathcal{H}} z_{iks} = 1 \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \tag{6}$$

$$\sum_{k \in \mathcal{H}} z_{jks} = 1 \quad \forall j \in \mathcal{J}, s \in \mathcal{S} \tag{7}$$

$$z_{iks} \leq x_k + y_{ks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{8}$$

$$z_{jks} \leq x_k + y_{ks} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{9}$$

$$z_{iks} \leq \sum_{m \in \mathcal{M}} a_{ikms} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \tag{10}$$

$$z_{jks} \leq \sum_{m \in \mathcal{M}} a_{jkms} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \tag{11}$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H}, m \in \mathcal{M}, s \in \mathcal{S} \tag{12}$$

$$\begin{aligned}
\sum_{i \in \mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{khms} \leq q_k &+ \sum_{j \in \mathcal{J}} O_{js} z_{jks} \\
&+ \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S}
\end{aligned} \tag{13}$$

$$q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \tag{14}$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \tag{15}$$

$$q_k, f_{khms} \geq 0 \quad \forall k, h, m, s \tag{16}$$

Constraints (3) define the hub establishment. For planned hubs, inventory is bounded by physical capacity (4), while the reactive hubs are required to be placed within safe zones (5). Constraints (6)-(11) rigorously enforce the single-allocation property to active hubs via available links. Constraint (12) restricts flow to be trans-shipped on intact paths. Constraints (13) and (14) maintain flow conservation and capacity limits. Specifically, (13) ensures that total supply, comprising inventory (q_k), external origins (O_{js}), and incoming flows – satisfies both victim demand (converted by γ) and outgoing trans-shipments. Variable domains are defined in (15)-(16).

3 Proposed Algorithm: PB-NSGA-II

The MO-IHLNDP is NP-hard due to its combinatorial structure and nonlinear Z_2. Exact solvers are intractable beyond small instances. We therefore propose the **Priority-Based NSGA-II (PB-NSGA-II)**. It integrates a custom decoding heuristic into the NSGA-II framework [6].

The key design principle is *implicit encoding*. Only first-stage strategic variables are encoded in the chromosome. All second-stage variables are reconstructed deterministically by the decoder for each scenario $s \in \mathcal{S}$. This drastically reduces the search space from $O(|\mathcal{S}| \times |\mathcal{V}|)$ to $O(|\mathcal{H}|)$.

3.1 Chromosome Encoding

Each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{W})$ encodes three segments:

- $\mathbf{X} \in \{0,1\}^{|\mathcal{H}|}$: binary hub activation vector. $X_k = 1$ if and only if hub k is planned before the disaster.
- $\mathbf{R} \in [0,1]^{|\mathcal{H}|}$: inventory ratio vector. The absolute pre-positioned inventory is decoded as:
$$q_k = R_k \cdot \frac{\hat{D}}{n_{\text{open}}}, \quad n_{\text{open}} = \textstyle\sum_k X_k,$$
where $\hat{D} = \gamma \cdot \max_s \sum_i D_{is}$ is the worst-case total demand in kg. This encoding ensures complete coverage. When $R_k = 1$ for all open hubs, the combined inventory exactly covers the worst-case demand. Capacity κ_k acts as a soft upper bound.
- $\mathbf{W} = (w_1, w_2, w_3) \in [0,1]^3$: heuristic weight vector. It governs the priority scoring during decoding.

3.2 Priority-Based Decoding Heuristic

The decoder is invoked for each scenario s. It proceeds through four steps:

Step 1 – Hub Activation. Planned hubs ($X_k = 1$) with risk $r_{ks} \leq \chi$ are activated. If no planned hub is safe, the hub with the lowest risk is activated reactively. Additional reactive hubs ($y_{ks} = 1$) may be activated later for unserviceable demand nodes.

Step 2 – Priority-Based Demand Assignment. Each demand node i is assigned a priority score:

$$\text{Score}_i = w_1 \cdot (\lambda_{is} \cdot D_{is}) - w_2 \cdot \min_{k \in \text{active}} \tau_{ikm^*},$$

where $m^* = \arg\min_m \tau_{ikm}$ subject to $a_{ikms} = 1$. Demands are sorted descendingly. They are assigned greedily to the highest-scoring active hub reachable via at least one mode. If no active hub can reach demand i, a reactive hub is opened from among safe candidates. Failing that, a constraint violation (CV) penalty of γD_{is} is incurred.

Step 3 – Origin Assignment and Flow Balancing. Each origin j is routed to the nearest active hub. A greedy trans-shipment loop then iteratively routes relief from surplus hubs to deficit hubs. It uses the cheapest available mode and incorporates the economies-of-scale discount α. Remaining deficits contribute proportionally to the CV score.

Step 4 – Objective Evaluation. Z_1 and Z_2 are accumulated as weighted expectations over scenarios using Equations (1) and (2). The deprivation waiting time is $\Omega_{is} = \tau_{ks} + 2\tau_{ikm^*}$. It is capped to prevent numerical overflow.

3.3 Genetic Operators and Selection

The mixed-type chromosome requires mode-specific operators. *Selection* uses a binary tournament on rank and crowding distance. *Crossover* ($p_c = 0.9$) utilizes Uniform Crossover for $\mathbf{X}$ and Simulated Binary Crossover (SBX, $\eta_c = 20$) for $\mathbf{R}$ and $\mathbf{W}$. *Mutation* ($p_m \propto 1/|\mathbf{C}|$) applies Bit-flip for $\mathbf{X}$ and Polynomial mutation ($\eta_m = 20$) for $\mathbf{R}$ and $\mathbf{W}$.

Constraint handling follows the constrained-dominance principle. Infeasible solutions (CV > 0) are always dominated by feasible ones. Among infeasible solutions, a lower CV dominates. The Pareto archive uses fast non-dominated sorting with crowding distance tie-breaking.

4 Computational Experiments

4.1 Experimental Setup

Implementation. PB-NSGA-II is implemented in C++17. Data generation and result analysis are performed in Python 3. All experiments run on a standard workstation (Intel Core i7, 16 GB RAM).

Algorithm parameters. Population $N = 100$; generations $G = 200$; $p_c = 0.9$; $\eta_c = \eta_m = 20$; $p_m = 1/|\mathbf{C}|$. The safety threshold is $\chi = 0.7$; discount $\alpha = 0.6$; $\gamma = 3.0$ kg/person; $\lambda_0 = 0.8$.

Performance metric. We use the **Hypervolume (HV)** indicator. The reference point is set to $1.1 \times \max(Z_1, Z_2)$ of the combined Pareto front. A larger HV indicates better coverage and spread of the Pareto front.

4.2 Dataset Description

Benchmark datasets. We adapt two widely-used hub location benchmarks: the **AP** set [7] (Euclidean instances of sizes 10–100 nodes) and **TR81** [11] (Turkish inter-city matrix, 81 nodes). Each benchmark is transformed into a DRND instance. We assign hub/origin/demand roles and generate three flood scenarios (mild, severe, extreme) with probabilities $(\pi_1, \pi_2, \pi_3) = (0.60, 0.30, 0.10)$. Supply and demand are calibrated to 1.5–2.5× total demand coverage. Road distances are normalized to a 500 km scale. Three transport modes operate concurrently (Table 1).

Table 1. Transport mode parameters.

Mode	Vehicle	Speed (km/h)	Cost ($/km)	Capacity (pers.)	Disruption
0	Truck	35	2.0	60	Road-dependent
1	Motorboat	25	5.0	25	Low (flood-resilient)
2	Helicopter	150	40.0	10	None (always available)

Central Vietnam case study. We generate two synthetic instances modelling the flood-prone **Vu Gia** – **Thu Bon** river basin (Da Nang – Quang Nam region):

- **CV-Small**: $|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$ – small-scale planning instance.
- **CV-Large**: $|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$ – high-resolution regional instance.

Node coordinates are drawn from real geographic locations. Road travel times are estimated via the OSRM API with a terrain-adjusted cost factor for inland areas. Scenario epicenters are sampled from flood-prone sub-regions. Risk indices decay with distance from the epicenter. Hub capacities are calibrated post-scenario generation to ensure structural feasibility ($\kappa_k \geq 3\hat{D}/|\mathcal{H}|$).

4.3 Results

Table 2 reports the final-generation hypervolume for each instance. For benchmark instances, we run one seed ($G = 200$, $N = 100$). For CV-Small, we run three independent seeds to assess robustness.

Table 2. Hypervolume (HV) results across all instances. CV-Small reports mean over 3 seeds.

Category	Instance	#Pareto pts	HV
AP Benchmark	AP10	100	1.17×10^{14}
	AP20	100	6.77×10^{14}
	AP25	100	2.20×10^{13}
	AP40	100	9.87×10^{13}
	AP50	100	4.94×10^{13}
	AP100	25	2.65×10^{14}
TR Benchmark	TR81	2	3.20×10^{13}
Case Study	CV-Small ($n = 3$)	152 (combined)	1.01×10^{12}
	CV-Large	100	3.23×10^{13}

Benchmark observations. The algorithm consistently produces well-spread Pareto fronts across all AP instances. AP20 yields the largest HV. This reflects the favourable trade-off structure of mid-size instances. AP100 produces fewer non-dominated points (25). This is expected as the search landscape becomes harder and population diversity is consumed. TR81 converges to only 2 non-dominated solutions. It indicates a strong dominance structure where a single hub configuration dominates most others in a dense graph.

Case study observations. For CV-Small, a combined Pareto front of 152 points demonstrates excellent solution diversity and reproducibility. CV-Large achieves 100 non-dominated solutions with an HV comparable to TR81. This illustrates that PB-NSGA-II scales effectively to large real-world instances. The smaller HV relative to AP benchmarks is attributable to the narrow feasible region. The flood disruption model severely restricts options (e.g., only 3 out of 5 hubs remain safe in extreme scenarios).

4.4 Managerial Insights

Analyzing the Pareto front of CV-Small reveals a clear convex trade-off. Reducing Z_2 (deprivation) by 30% requires increasing Z_1 (logistics cost) by approximately 3×. This reflects the exponential structure of the deprivation function. Reducing waiting times for high-risk isolated areas is disproportionately costly.

Network topology analysis further highlights structural shifts. Under the *mild* scenario, the optimal strategy uses road transport from all 5 candidate hubs with moderate inventory pre-positioning. Under the *extreme* scenario, 2 hubs become unsafe ($r_{ks} > \chi$). The network dynamically switches to water and air transport. The remaining active hubs receive larger inventory pre-commitments. This is compensated by increased lateral trans-shipment. This shift confirms the value of the two-stage stochastic formulation. A deterministic single-scenario model would systematically under-invest in network resilience.

5 Conclusion and Future Work

We proposed PB-NSGA-II for the Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP). The model incorporates multi-modal transport, stochastic flood scenarios, deprivation cost, and Daganzo's continuous approximation. Our priority-based decoding scheme efficiently handles second-stage variables and maintains feasibility in disrupted networks. Experiments on AP and TR81 benchmarks validate the algorithm's effectiveness. The Central Vietnam case study demonstrates its practical applicability.

Limitations. The current model relies on a fixed set of pre-defined scenarios. The decoding heuristic is greedy and may miss globally optimal demand assignments.

Future work will explore exact branch-and-cut approaches for small instances to establish lower bounds. We also aim to incorporate real-time sensor data via online re-optimization and extend the framework to multi-period relief operations.

References

1. Altay, N., Green III, W.G.: Or/ms research in disaster operations management. European journal of operational research **175**(1), 475–493 (2006)
2. ASEAN Secretariat: Asean disaster resilience outlook: Preparing for a future beyond 2025. Tech. rep., ASEAN Coordinating Centre for Humanitarian Assistance on disaster management (AHA Centre), Jakarta (2021), https://asean.org/
3. Boonmee, C., Arimura, M., Asada, T.: Facility location optimization model for emergency humanitarian logistics. International journal of disaster risk reduction **24**, 485–498 (2017)
4. Center for Disaster Philanthropy: 2025 southeast asia severe storms (December 2025), https://disasterphilanthropy.org/disasters/2025-southeast-asia-severe-storms/
5. Daganzo, C.F.: Logistics systems analysis. Springer (2005)
6. Deb, K., Pratap, A., Agarwal, S., Meyarivan, T.: A fast and elitist multiobjective genetic algorithm: Nsga-ii. IEEE transactions on evolutionary computation **6**(2), 182–197 (2002)
7. Ernst, A.T., Krishnamoorthy, M.: Efficient algorithms for the uncapacitated single allocation p-hub median problem. Location science **4**(3), 139–154 (1996)
8. Hasani, A., Mokhtari, H.: An integrated relief network design model under uncertainty: A case of iran. Safety Science **111**, 22–36 (2019)
9. Holguín-Veras, J., Pérez, N., Jaller, M., Van Wassenhove, L.N., Aros-Vera, F.: On the appropriate objective function for post-disaster humanitarian logistics models. Journal of Operations Management **31**(5), 262–280 (2013)
10. IFRC: Vietnam: Floods - final report, operation n° mdrvn020. ReliefWeb (April 2022), https://reliefweb.int/report/viet-nam/vietnam-floods-final-report-operation-n-mdrvn020, report detailing the destruction of over 165 km of national highways and isolation of central provinces.
11. Kara, B.Y., Tansel, B.C.: On the single-assignment p-hub center problem. European Journal of Operational Research **125**(3), 648–655 (2000)
12. Kiet, A.: Railway damage accounted for us$1.9 million after severe flood in central vietnam. The Hanoi Times (December 2025), https://hanoitimes.vn/railway-damage-accounted-for-ususd1-9-million-after-severe-flood-in-central-vietnam.925847.html
13. Tofighi, S., Torabi, S.A., Mansouri, S.A.: Humanitarian logistics network design under mixed uncertainty. European journal of operational research **250**(1), 239–250 (2016)
14. Van Wassenhove, L.N.: Humanitarian aid logistics: supply chain management in high gear. Journal of the Operational research Society **57**(5), 475–489 (2006)
15. Voice of Vietnam: (translated) natural disasters in 2025: Fierce, abnormal, causing heavy losses in humans and property (December 2025), https://vov.vn/xa-hoi/thien-tai-nam-2025-khoc-liet-di-thuong-gay-thiet-hai-nang-ne-ve-nguoi-va-cua-post1258134.vov, vOV.VN
16. Yahyaei, M., Bozorgi-Amiri, A.: Robust reliable humanitarian relief network design: an integration of shelter and supply facility location. Annals of Operations Research **283**, 897–916 (2019)